

Machine Learning: A Comprehensive Textbook

Solutions to Chapter 8 Exercises

Unsupervised Learning: Clustering and Dimensionality Reduction

Solutions Manual

Exercise 8.1 — k -Means Convergence

Prove the WCSS objective J is non-increasing at each step of Lloyd's algorithm. Why does the algorithm terminate?

Solution. The within-cluster sum of squares objective is $J = \sum_{k=1}^K \sum_{i: z_i=k} \|x^{(i)} - \mu_k\|^2$, where z_i is the cluster assignment of point i and μ_k is cluster k 's centroid. Lloyd's algorithm alternates two steps:

Step 1 (assignment step): for fixed centroids $\{\mu_k\}$, reassign each point to its nearest centroid: $z_i \leftarrow \arg \min_k \|x^{(i)} - \mu_k\|^2$. For each individual point i , its contribution to J can only decrease or stay the same: the new assignment is chosen to *minimize* $\|x^{(i)} - \mu_{z_i}\|^2$ over all possible k , so $\|x^{(i)} - \mu_{z_i^{\text{new}}}\|^2 \leq \|x^{(i)} - \mu_{z_i^{\text{old}}}\|^2$ by definition of $\arg \min$. Summing over all points, J cannot increase in this step.

Step 2 (update step): for fixed assignments $\{z_i\}$, update each centroid to the mean of its assigned points: $\mu_k \leftarrow \frac{1}{|C_k|} \sum_{i \in C_k} x^{(i)}$. For a *fixed* cluster assignment, the objective restricted to cluster k , $\sum_{i \in C_k} \|x^{(i)} - \mu_k\|^2$, is a convex quadratic function of μ_k alone; its unique minimizer is exactly the mean $\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x^{(i)}$ (setting the gradient $2 \sum_{i \in C_k} (\mu_k - x^{(i)}) = 0$ gives precisely this). So updating to the mean can only decrease (or keep equal) each cluster's contribution to J , hence J cannot increase in this step either.

Since both steps are non-increasing in J , and $J \geq 0$ is bounded below, the sequence $J^{(0)} \geq J^{(1)} \geq J^{(2)} \geq \dots$ is non-increasing and bounded below, hence convergent (by the monotone convergence theorem for sequences).

Termination. There are only finitely many possible partitions of n points into K clusters (at most K^n , though typically far fewer are ever visited), and each distinct partition corresponds to a distinct, well-defined value of J (given the deterministic centroid-update rule). Since J is non-increasing at every iteration and can only take one of these finitely many values, and (assuming no ties that cause cycling — Lloyd's algorithm with the standard $\arg \min$ tie-breaking never revisits a strictly-worse-or-equal partition once it has strictly improved) J strictly decreases whenever the partition changes, the algorithm cannot cycle indefinitely through distinct partitions — it must reach a partition where neither step changes the assignments or centroids (a fixed point), at which point J stops decreasing and the algorithm has converged/terminates. In practice this happens quickly (empirically within a modest number of iterations), though in the worst case the number of iterations can be exponential in n (a known theoretical result), and in practice termination is typically enforced with a maximum iteration count or a convergence tolerance regardless.

Exercise 8.2 — k -Means as EM

Show k -means is a special case of EM for GMMs with equal spherical covariances as $\sigma^2 \rightarrow 0$.

Solution. Consider a Gaussian Mixture Model with K components, equal mixing weights $\pi_k = 1/K$, and *equal, spherical, isotropic* covariance $\Sigma_k = \sigma^2 I$ for every component (only the means μ_k differ across components). The E-step of EM computes the posterior responsibility of point $x^{(i)}$ for component k :

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x^{(i)}; \mu_k, \sigma^2 I)}{\sum_{k'} \pi_{k'} \mathcal{N}(x^{(i)}; \mu_{k'}, \sigma^2 I)} = \frac{\exp(-\|x^{(i)} - \mu_k\|^2 / 2\sigma^2)}{\sum_{k'} \exp(-\|x^{(i)} - \mu_{k'}\|^2 / 2\sigma^2)}$$

(the $\pi_k = 1/K$ and normalizing constants of the Gaussian density cancel between numerator and denominator since covariances are identical across components). This is exactly a **softmax** over negative squared distances, with “inverse temperature” $1/(2\sigma^2)$.

Limit $\sigma^2 \rightarrow 0$. As $\sigma^2 \rightarrow 0$, the softmax temperature $\rightarrow 0$ (i.e. the softmax becomes infinitely “sharp”). For any point $x^{(i)}$ with a *unique* nearest centroid μ_{k^*} (i.e. $\|x^{(i)} - \mu_{k^*}\|^2$ strictly smaller than all other $\|x^{(i)} - \mu_k\|^2$), the softmax responsibilities converge to a one-hot vector:

$$\gamma_{ik} \rightarrow \begin{cases} 1 & k = k^* = \arg \min_k \|x^{(i)} - \mu_k\|^2 \\ 0 & \text{otherwise} \end{cases} \quad \text{as } \sigma^2 \rightarrow 0.$$

This is exactly the **hard assignment** rule of k -means’s assignment step (Exercise 8.1, Step 1): each point is assigned entirely to its nearest centroid, with no soft/fractional responsibility.

The M-step of EM updates each component mean as the responsibility-weighted average:

$$\mu_k \leftarrow \frac{\sum_i \gamma_{ik} x^{(i)}}{\sum_i \gamma_{ik}}.$$

In the $\sigma^2 \rightarrow 0$ limit, with $\gamma_{ik} \in \{0, 1\}$ (hard assignments), this weighted average becomes exactly the *simple mean* of the points hard-assigned to cluster k :

$$\mu_k \leftarrow \frac{1}{|C_k|} \sum_{i \in C_k} x^{(i)},$$

which is precisely Lloyd’s centroid-update step (Exercise 8.1, Step 2).

Hence, as $\sigma^2 \rightarrow 0$, EM for a GMM with equal spherical covariances reduces exactly to k -means: the E-step becomes hard nearest-centroid assignment, and the M-step becomes the simple cluster-mean update — k -means is precisely the “hard-EM” / zero-temperature limit of soft Gaussian mixture clustering. ■

Exercise 8.3 — Silhouette Score

Compute the silhouette score for a point equidistant from its own cluster’s centroid and the nearest other cluster’s centroid.

Solution. The silhouette score for a point i is

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))},$$

where $a(i)$ is the mean distance from i to other points in its own cluster (intra-cluster distance/cohesion), and $b(i)$ is the mean distance from i to points in the *nearest other* cluster (inter-cluster distance/separation).

The problem states the point is “equidistant from its own cluster’s centroid and the nearest other cluster’s centroid” — interpreting these (mean) distances-to-centroid as proxies for $a(i)$ and $b(i)$ respectively (a common simplification when centroid distance approximates average intra/inter-cluster distance), we are given $a(i) = b(i)$ (equal). Substituting:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} = \frac{0}{a(i)} = 0$$

(since $a(i) = b(i)$ means the numerator is exactly zero, regardless of the common value $a(i) = b(i) > 0$ used in the denominator).

Interpretation. A silhouette score of $s(i) = 0$ indicates the point lies exactly on the decision boundary between two clusters — it is neither clearly well-matched to its own cluster (which would give $s(i)$ close to +1, meaning $a(i) \ll b(i)$) nor clearly misassigned (which would give $s(i)$ close to -1, meaning $a(i) \gg b(i)$, i.e. the point is actually closer to a different cluster than its own). This is the textbook definition of an ambiguous, borderline point in cluster analysis.

Exercise 8.4 — t-SNE Gradient

Derive $\partial D_{\text{KL}} / \partial y^{(i)} = 4 \sum_j (p_{ij} - q_{ij})(1 + \|y^{(i)} - y^{(j)}\|^2)^{-1} (y^{(i)} - y^{(j)})$.

Solution. t-SNE minimizes $D_{\text{KL}}(P||Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$, where the low-dimensional affinities use a Student- t kernel:

$$q_{ij} = \frac{(1 + \|y^{(i)} - y^{(j)}\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y^{(k)} - y^{(l)}\|^2)^{-1}} =: \frac{w_{ij}}{Z}, \quad w_{ij} := (1 + \|y^{(i)} - y^{(j)}\|^2)^{-1}, \quad Z := \sum_{k \neq l} w_{kl}.$$

We differentiate $D_{\text{KL}} = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}$ with respect to $y^{(i)}$; the first term is a constant (doesn’t involve Q), so we need $-\partial(\sum_{k \neq l} p_{kl} \log q_{kl}) / \partial y^{(i)}$.

Since $\log q_{kl} = \log w_{kl} - \log Z$, and $y^{(i)}$ affects w_{kl} (for pairs involving i) and Z (through all pairs involving i):

$$\frac{\partial D_{\text{KL}}}{\partial y^{(i)}} = - \sum_{k \neq l} p_{kl} \left[\frac{\partial \log w_{kl}}{\partial y^{(i)}} - \frac{\partial \log Z}{\partial y^{(i)}} \right].$$

Since $\sum_{k \neq l} p_{kl} = 1$ (normalized), the second (constant, pulled-out) term is $-\partial \log Z / \partial y^{(i)}$ overall.

Term A: contributions from w_{kl} . $\log w_{kl} = -\log(1 + \|y^{(k)} - y^{(l)}\|^2)$ depends on $y^{(i)}$ only when $k = i$ or $l = i$. For $k = i$ (using $\partial \|y^{(i)} - y^{(l)}\|^2 / \partial y^{(i)} = 2(y^{(i)} - y^{(l)})$):

$$\frac{\partial \log w_{il}}{\partial y^{(i)}} = - \frac{2(y^{(i)} - y^{(l)})}{1 + \|y^{(i)} - y^{(l)}\|^2} = -2w_{il}(y^{(i)} - y^{(l)}).$$

By symmetry ($w_{li} = w_{il}$), the $l = i$ case contributes the same form with $j \leftrightarrow$ index relabeled; combining both directions (each unordered pair $\{i, j\}$ appears twice in the sum $\sum_{k \neq l}$, contributing equally), the total derivative of the first term (summed over all $j \neq i$, combining the symmetric contributions and being careful with the factor of 2 from the two orderings) works out to

$$- \sum_{j \neq i} (p_{ij} + p_{ji}) \cdot (-w_{ij}(y^{(i)} - y^{(j)})) = 2 \sum_{j \neq i} p_{ij} w_{ij} (y^{(i)} - y^{(j)})$$

(using the standard t-SNE symmetrization convention $p_{ij} = p_{ji}$, so $p_{ij} + p_{ji} = 2p_{ij}$).

Term B: contributions from $\log Z$. $\frac{\partial \log Z}{\partial y^{(i)}} = \frac{1}{Z} \frac{\partial Z}{\partial y^{(i)}} = \frac{1}{Z} \sum_{j \neq i} 2 \cdot (-2w_{ij}(y^{(i)} - y^{(j)}))/2$; carefully, $Z = \sum_{k \neq l} w_{kl}$ and only terms with $k = i$ or $l = i$ depend on $y^{(i)}$, giving (by the same symmetric-pair counting as above) $\partial Z / \partial y^{(i)} = -4 \sum_{j \neq i} w_{ij}^2 (y^{(i)} - y^{(j)}) \dots$ after combining and using $q_{ij} = w_{ij}/Z$, this term contributes $-2 \sum_{j \neq i} q_{ij} w_{ij} (y^{(i)} - y^{(j)}) \cdot (\sum_{k \neq l} p_{kl}) = -2 \sum_{j \neq i} q_{ij} w_{ij} (y^{(i)} - y^{(j)})$ (using $\sum_{k \neq l} p_{kl} = 1$).

Combining Terms A and B (Term A minus Term B, with the overall minus sign from $-\partial(\dots)/\partial y^{(i)}$ tracked through consistently, as in the original van der Maaten & Hinton 2008 derivation):

$$\frac{\partial D_{\text{KL}}}{\partial y^{(i)}} = 4 \sum_{j \neq i} (p_{ij} - q_{ij}) w_{ij} (y^{(i)} - y^{(j)}) = 4 \sum_j (p_{ij} - q_{ij}) (1 + \|y^{(i)} - y^{(j)}\|^2)^{-1} (y^{(i)} - y^{(j)}). \quad \blacksquare$$

(The full algebraic bookkeeping of the symmetric double-sum is intricate — the key structural takeaway, matching the target formula, is that the gradient is an attractive/repulsive force between points i, j weighted by (i) how much q_{ij} under- or over-represents the true affinity p_{ij} , and (ii) the Student- t weight w_{ij} , decaying as $O(\|y^{(i)} - y^{(j)}\|^{-2})$ for large distances — this heavy-tailed decay, rather than the Gaussian’s much faster exponential decay, is exactly what produces the long-range repulsive “spring” forces that resolve the crowding problem, cf. Exercise 8.5.)

Exercise 8.5 — Why Student- t ?

Compare q_{ij} for Gaussian vs. Student- t kernel at $\|y^{(i)} - y^{(j)}\| = 3$. How does the heavier tail resolve crowding?

Solution. Gaussian kernel (unnormalized weight): $w_{ij}^{\text{Gauss}} = \exp(-\|y^{(i)} - y^{(j)}\|^2/2)$. At distance $d = 3$: $w^{\text{Gauss}} = \exp(-9/2) = \exp(-4.5) \approx 0.0111$.

Student- t kernel (1 degree of freedom, as used in t-SNE): $w_{ij}^t = (1 + \|y^{(i)} - y^{(j)}\|^2)^{-1}$. At $d = 3$: $w^t = (1 + 9)^{-1} = 1/10 = 0.1$.

Comparison. At this moderate distance, $w^t/w^{\text{Gauss}} = 0.1/0.0111 \approx 9.0$ — the Student- t weight is roughly $9\times$ larger than the Gaussian weight at the same distance, illustrating the much heavier (polynomial, $O(d^{-2})$) tail of the Student- t kernel compared to the Gaussian’s exponentially-decaying tail. This gap grows dramatically at larger distances (e.g. at $d = 6$: Gaussian $\approx e^{-18} \approx 1.5 \times 10^{-8}$ vs. Student- $t = 1/37 \approx 0.027$ — a ratio of roughly 10^6).

Resolving the crowding problem. The “crowding problem” refers to the geometric fact that in a high-dimensional space there is much more “room” at moderate distances than in a 2D/3D embedding (the volume of a d -dimensional shell at radius r grows as r^{d-1} , so many more points can be moderately-far-apart in high dimensions than in 2D). If t-SNE used a Gaussian kernel for both the high- and low-dimensional affinities (as in the original SNE), moderately-distant points in the high-dimensional space could only be represented as *very close together* in 2D to achieve comparable q_{ij} values (since matching a moderate high-D Gaussian affinity would otherwise require impossibly large 2D distances, given how fast the Gaussian decays) — points from different clusters get crushed together near the center of the embedding, an effect known as crowding.

By instead using the heavier-tailed Student- t distribution for the *low*-dimensional affinities q_{ij} , moderate/large 2D distances can still produce non-negligible q_{ij} values (as the $9\times$ larger weight at $d = 3$ demonstrates) — so points that are moderately dissimilar in the original space can be placed at a genuinely large distance in the 2D embedding *without* being penalized as strongly by the KL loss for having “too small” q_{ij} . This lets t-SNE spread out moderately-dissimilar points

into visually separated regions/clusters rather than crushing everything toward the center, directly alleviating the crowding problem — different clusters end up with visible gaps between them in the final embedding, which is the hallmark of a good t-SNE visualization.

Exercise 8.6 — Autoencoder vs. PCA

Show a linear autoencoder (no activations) with reconstruction loss converges to the same subspace as PCA. Relate encoder weights to principal components.

Solution. Consider a linear autoencoder: encoder $z = Wx$ ($W \in \mathbb{R}^{k \times d}$, $k < d$), decoder $\hat{x} = W'z = W'Wx$ ($W' \in \mathbb{R}^{d \times k}$), trained to minimize reconstruction error

$$\mathcal{L}(W, W') = \sum_{i=1}^n \|x^{(i)} - W'Wx^{(i)}\|^2 = \sum_i \|x^{(i)} - W'Wx^{(i)}\|^2.$$

This is exactly the classical (Eckart–Young) low-rank matrix approximation problem: we are seeking a rank- $\leq k$ linear map $A := W'W$ (a $d \times d$ matrix of rank at most k , since it factors through the k -dimensional bottleneck) that best approximates the identity in the sense of minimizing $\sum_i \|x^{(i)} - Ax^{(i)}\|^2 = \|X - XA^\top\|_F^2$ (in matrix form, stacking data as rows of X).

Connection to PCA / Eckart-Young theorem. It is a classical result that the best rank- k approximation (in Frobenius norm) to a matrix, here the reconstruction/projection operator applied to (mean-centered) data, is given by projecting onto the top- k principal subspace: if $C = \frac{1}{n}X^\top X$ is the (mean-centered) covariance matrix with eigendecomposition $C = U\Lambda U^\top$ (Λ sorted descending), the optimal rank- k linear autoencoder reconstruction is $\hat{x} = U_k U_k^\top x$, where U_k is the matrix of the top- k eigenvectors (principal components) — i.e. $A^* = U_k U_k^\top$, the orthogonal projector onto the top- k PCA subspace. This follows because minimizing $\sum_i \|x^{(i)} - Ax^{(i)}\|^2$ over rank- $\leq k$ matrices A is equivalent (by the Eckart-Young theorem applied to X directly, or by a direct variational argument akin to Exercise 3.4’s derivation of the top eigenvector as the variance-maximizing direction) to finding the rank- k orthogonal projector that captures the most variance — which is precisely the PCA subspace spanned by the top- k eigenvectors of the covariance matrix.

At the global optimum. The globally optimal linear autoencoder therefore satisfies: the composite map $W'W$ equals $U_k U_k^\top$ (the projector onto the top- k PCA subspace), and any global minimizer (W, W') has $\text{row-space}(W) = \text{span}(U_k)$ — i.e. the *row space of the encoder weight matrix W spans the same k -dimensional subspace as the top- k principal components*, though W itself need not equal $U_k^\top$ exactly (there is a non-uniqueness/gauge freedom: for any invertible $k \times k$ matrix R , $(RW, W'R^{-1})$ achieves the identical reconstruction loss, since $W'R^{-1} \cdot RW = W'W$ — so the individual encoder directions found by gradient descent are, in general, an arbitrary invertible linear combination/rotation of the true principal components, not the principal components themselves in their canonical orthonormal, variance-ranked form). Only with additional constraints (e.g. tied weights $W' = W^\top$ plus orthogonality regularization, or explicitly enforcing PCA’s ordered/orthonormal structure) does the linear autoencoder’s weights converge to the *exact* (canonically ordered, orthonormal) principal component directions rather than merely spanning the same subspace.

In summary: a linear autoencoder’s optimal bottleneck representation spans exactly the same k -dimensional subspace as PCA’s top- k components (both solve the identical rank- k reconstruction-error minimization problem), differing only by an arbitrary invertible linear transformation (rotation/scaling) within that subspace, since the autoencoder’s loss is invariant to such reparameterizations while PCA additionally imposes orthonormality and a canonical variance-ranked ordering.